Exp.no : 14

FREQUENCY OF OCCURENCE

a) AIM:

Conduct an experiment to show data visualization using line plot

PYTHON CODE:

```
text = """
Data science is an important field of study.
Data science combines statistics, mathematics,
programming, and machine learning.
Data scientists collect data from different sources.
They clean data, analyze data, and visualize data.
Python is widely used in data science.
Python provides libraries like NumPy, Pandas,
Matplotlib, and Scikit-learn.
These libraries help process large datasets.
Data analysis helps organizations make decisions.
Machine learning uses data to build predictive models.
Data visualization presents information through graphs.
Graphs include line plots, bar charts, and pie charts.
Data mining discovers useful patterns in data.
Big data contains large amounts of information.
Data engineers prepare data for analysis.
Data analysts study trends and summarize results.
Artificial intelligence uses data to learn patterns.
Good data improves the accuracy of predictions.
Data science is useful in healthcare, education,
banking, business, agriculture, and transportation.
Students learn data science to solve real-world problems.
The demand for data science professionals is increasing.
Data processing is an essential step in every project.
"""

word = input("Enter the word to count: ").lower()

words = text.lower().replace(".", "").replace(",", "").split()
frequency = words.count(word)

print("\nTotal words in the text:", len(words))
print("Word searched:", word)
print("Frequency of occurrence:", frequency)
```

OUTPUT:

Enter the word to count: data

Total words in the text: 131
Word searched: data
Frequency of occurrence: 15

RESULT:

The code is executed successfully.